Academic year 2026 - 2027

Bachelor in Mathematics, option: Data Science

Project / Internship Title

Fraud Detection Using Machine Learning

Student Name

Carla Radieddine

Supervisor: Dr. Abbas Rammal

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Abstract

Fraud detection is an important application of data science, especially in financial systems where fraudulent transactions can cause financial loss and affect customer trust. The goal of this project is to identify fraudulent transactions by applying machine learning techniques to a large transaction dataset from Kaggle.

In order to preprocess the dataset, missing values were initially handled, categorical variables were encoded, numerical features were scaled, and additional features including transaction hour, day, weekday, and client age were created. Logistic Regression, Random Forest, XGBoost, Isolation Forest, and One-Class SVM were among the supervised and unsupervised machine learning models that were implemented and evaluated. Due to the highly imbalanced nature of the dataset, the models were evaluated using appropriate evaluation metrics such as precision, recall, F1-score, ROC-AUC and confusion matrices rather than just accuracy.

Results showed that XGBoost performed the best of all machine learning approaches, with a good balance between precision and recall. Random Forest achieved the highest recall and the lowest estimated business cost, making it useful when the main objective is to minimize missed fraud cases. PCA was also implemented to study the impact of dimensionality reduction, but it did not improve model performance. Feature importance analysis showed that transaction category and transaction time were among the most influential indicators of fraud.

Overall, this project demonstrates that advanced supervised learning models are effective for fraud detection when combined with proper preprocessing, suitable evaluation metrics and business-oriented analysis.

Key words: Fraud detection, machine learning, imbalanced classification, XGBoost, Random Forest, PCA, feature importance, business cost evaluation.

1 Introduction

Fraud detection is an important problem in financial systems because fraudulent transactions can lead to financial losses, customer dissatisfaction and reduced trust in digital payment platforms. As online payments and electronic banking continue to grow, financial institutions need efficient methods to identify suspicious transactions quickly and accurately.

Traditional fraud detection methods often rely on fixed rules or manual review. However, these methods may not be effective when fraud patterns become more complex and change over time. Machine learning provides a more flexible approach because it can learn patterns from historical transaction data and use them to classify new transactions as fraudulent or non-fraudulent.

This project focuses on the development and evaluation of machine learning models for fraud detection using a large transaction dataset. The main objective is to compare different supervised and unsupervised learning approaches and determine which model performs best in identifying fraudulent transactions. Since fraud cases are rare compared to normal transactions, the project also considers the problem of class imbalance and uses evaluation metrics such as precision, recall, F1-score, ROC-AUC and confusion matrices.

In addition to traditional model evaluation, this project includes dimensionality reduction using PCA, feature importance analysis, cross-validation and a business cost-based evaluation. These steps provide a more complete understanding of model performance from both technical and practical perspectives.

2 Problem Definition

The problem addressed in this project is the detection of fraudulent financial transactions using machine learning. Each transaction must be classified as either non-fraudulent or fraudulent based on its characteristics, such as transaction amount, category, location, time and customer-related information.

This is a binary classification problem, where the target variable has two possible classes: 0 for non-fraudulent transactions and 1 for fraudulent transactions. The main challenge is that fraudulent transactions represent only a very small portion of the dataset, making the data highly imbalanced.

In fraud detection, identifying fraudulent transactions is more important than achieving high overall accuracy. A model that predicts most transactions as non-fraudulent may still achieve high accuracy, but it would fail to detect actual fraud cases. Therefore, this project focuses on evaluation metrics that are more suitable for imbalanced classification, such as precision, recall, F1-score, ROC-AUC and confusion matrices.

The goal of this project is to build and compare several machine learning models in order to identify the most effective approach for fraud detection. The project also considers the practical impact of false positives and false negatives through a business cost-based evaluation.

3 Dataset Description

The dataset used in this project was obtained from Kaggle and contains credit card transaction records used for fraud detection. Each record represents a transaction and includes information related to the transaction, customer, merchant, category, time, location and fraud status.

The target variable is is_fraud where 0 represents a non-fraudulent transaction and 1 represents a fraudulent transaction. The dataset contains more than one million transaction records, making it suitable for testing machine learning models on a large-scale fraud detection problem.

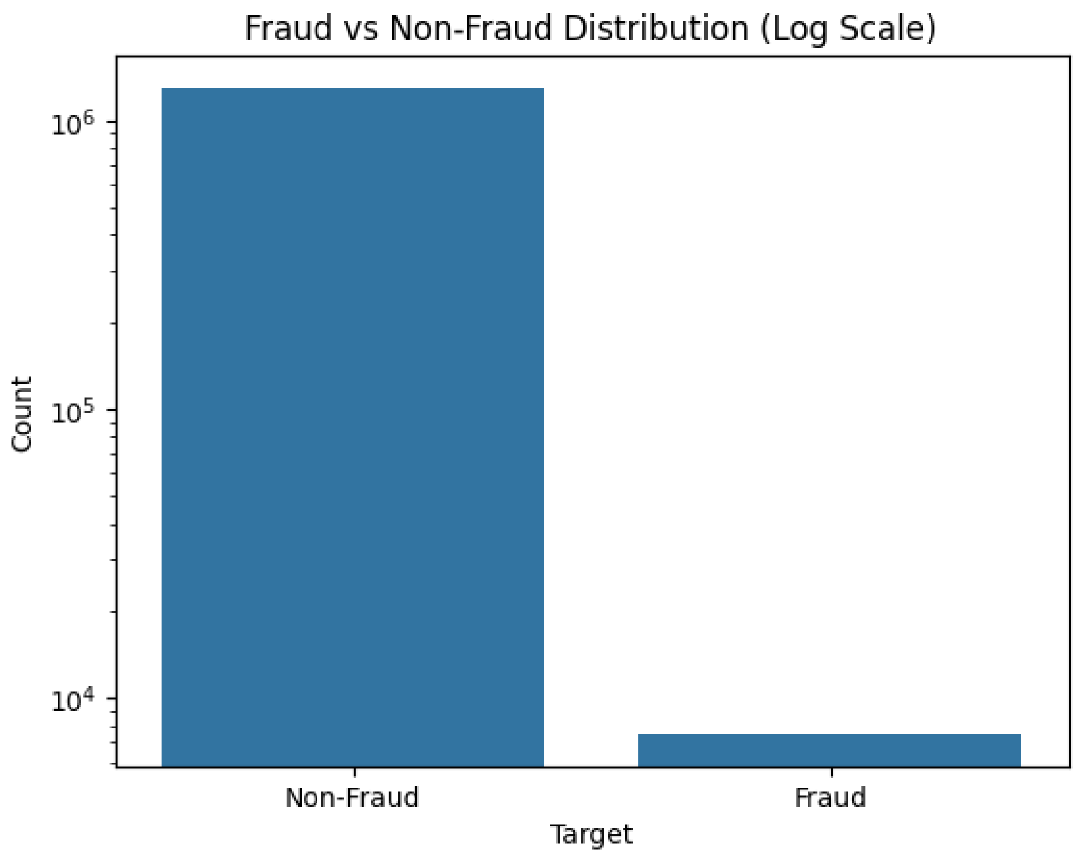

Figure 1: Fraud vs Non-Fraud Class Distribution (Log Scale)

The class distribution shows that the dataset is highly imbalanced, with non-fraudulent transactions representing the majority of the records and fraudulent transactions representing only a small minority. This imbalance is an important challenge because a model can achieve high accuracy by predicting most transactions as non-fraudulent, while still failing to detect actual fraud cases.

For this reason, accuracy alone is not sufficient to evaluate model performance. Metrics such as precision, recall, F1-score, ROC-AUC, and confusion matrices are more appropriate for assessing the ability of the models to detect fraud.

4 Methodology

This section describes the main steps followed to build the fraud detection system. The workflow includes data preprocessing, feature engineering, model training, evaluation, cross-validation, PCA implementation, feature importance analysis, and business cost-based evaluation.

4.1 Data Preprocessing

Data preprocessing was an essential step because the dataset contained different types of variables, including numerical, categorical and date-related features. Before training the models, unnecessary columns such as identifiers, personal information and raw date fields were removed to reduce noise and avoid using features that do not contribute meaningfully to fraud detection.

Missing values were handled using imputation techniques. Numerical features were filled using the median, while categorical features were filled using the most frequent value. These choices help preserve the structure of the dataset while avoiding errors during model training.

Categorical variables were converted into numerical format using one-hot encoding, since machine learning models cannot directly process text-based categories. Numerical features were scaled using standardization when required, especially for models such as Logistic Regression and One-Class SVM that are sensitive to feature scale.

4.2 Feature Engineering

Feature engineering was performed to create new variables that could help the models better identify fraud patterns. The transaction date and time were converted into useful time-based features, including the transaction hour, day and weekday. These features are important because fraudulent activity may occur more frequently during specific hours or days.

The customer’s age was also calculated using the customer’s date of birth. Age can be useful in fraud detection because transaction behavior may vary across different age groups.

After creating these features, the original raw date columns were removed to avoid redundancy. The final dataset used for modeling contained both the original relevant transaction features and the newly engineered features.

4.3 Preprocessing Pipeline

A preprocessing pipeline was created to ensure that all transformations were applied consistently to both training and testing data. The pipeline handled numerical and categorical features separately.

For numerical features, missing values were replaced using the median, and standardization was applied when needed. For categorical features, missing values were replaced using the most frequent category, and one-hot encoding was used to convert categories into numerical format.

These transformations were combined using a ColumnTransformer, allowing each type of feature to be processed appropriately. Using a pipeline also helps prevent data leakage because preprocessing steps are learned only from the training data and then applied to the test data.

4.4 Model Training

Several machine learning models were trained and evaluated in order to compare different approaches to fraud detection. The supervised models included Logistic Regression, Random Forest and XGBoost. These models were trained using labeled data, where each transaction was identified as fraudulent or non-fraudulent.

Logistic Regression was used as a simple baseline model. Random Forest was included as an ensemble tree-based model that can capture non-linear relationships. XGBoost was also implemented because it is a powerful gradient boosting algorithm that often performs well on structured tabular data.

In addition to supervised models, two unsupervised anomaly detection models were implemented: Isolation Forest and One-Class SVM. These models were used to compare traditional supervised fraud detection with anomaly detection techniques. Since these models output anomaly labels, their predictions were converted into the same format as the target variable, where 0 represents non-fraud and 1 represents fraud.

4.5 Evaluation Metrics

The models were evaluated using metrics that are suitable for imbalanced classification. Accuracy was calculated, but it was not considered sufficient on its own because the dataset contains far more non-fraudulent transactions than fraudulent transactions. A model could achieve high accuracy by predicting most transactions as non-fraudulent while still failing to detect fraud.

Precision was used to measure how many transactions predicted as fraudulent were actually fraud. Recall was used to measure how many actual fraud cases were correctly detected. F1-score was used to balance precision and recall, making it useful when both false positives and false negatives are important.

ROC-AUC was used for supervised models that provide probability outputs, such as Logistic Regression, Random Forest, and XGBoost. It measures the model's ability to distinguish between fraudulent and non-fraudulent transactions. Confusion matrices were also used to analyze true positives, true negatives, false positives, and false negatives in more detail.

For Isolation Forest and One-Class SVM, ROC-AUC was not included because these models were evaluated mainly using their binary anomaly predictions rather than probability outputs.

4.6 Cross-Validation

To improve the reliability of the model evaluation, 5-fold Stratified Cross-Validation was applied to the supervised models. Cross-validation allows the model to be trained and tested on different subsets of the data, which provides a more stable estimate of performance compared to a single train-test split.

Stratified K-Fold was used because the dataset is highly imbalanced. This ensures that each fold maintains approximately the same proportion of fraudulent and non-fraudulent transactions as the original dataset.

The use of cross-validation also helped confirm that the model results were not dependent on one specific random data split. This was especially important for comparing the stability of Logistic Regression and XGBoost.

4.7 PCA Implementation

Principal Component Analysis (PCA) was implemented to study the effect of dimensionality reduction on model performance. PCA reduces the number of features by transforming them into principal components while preserving as much information as possible.

In this project, PCA was applied only to numerical features after imputation and scaling. Categorical features were kept separate and encoded using one-hot encoding. This approach was used because applying PCA directly after one-hot encoding can create issues due to the sparse structure of encoded categorical variables.

After applying PCA, the selected models were retrained and evaluated to compare their performance before and after dimensionality reduction. This helped determine whether reducing the feature space improved or reduced the ability of the models to detect fraud.

4.8 Feature Importance Analysis

Feature importance analysis was performed using the best-performing supervised model, XGBoost. This step was used to identify which features contributed the most to the model's fraud detection decisions.

Feature importance helps improve the interpretability of the model by showing which transaction characteristics were most influential. In fraud detection, this is important because it allows the results to be explained beyond simply predicting whether a transaction is fraudulent or not.

The analysis focused on the original feature space rather than the PCA-transformed features, since PCA components are harder to interpret. This made it possible to understand the importance of meaningful variables such as transaction category, time, amount, and location-related features.

4.9 Business Cost-Based Evaluation

In addition to traditional machine learning metrics, a business cost-based evaluation was performed to assess the practical impact of model errors. In fraud detection, false negatives and false positives do not have the same consequences. A false negative means that a fraudulent transaction was missed, which may result in financial loss. A false positive means that a normal transaction was incorrectly flagged as fraud, which may cause customer inconvenience or require manual review.

To reflect this difference, false negatives were assigned a higher cost than false positives. The total estimated cost was calculated using the following formula:

Total Cost = (False Negatives × Cost of False Negative) + (False Positives × Cost of False Positive)

This evaluation provides a more realistic perspective on model performance by considering the financial impact of incorrect predictions. It also helps determine whether the best model based on machine learning metrics is also the best model from a business perspective.

5 Results and Discussion

This section presents and discusses the results obtained from the different fraud detection models. The analysis focuses on model performance, cross-validation stability, the effect of PCA, feature importance and business cost-based evaluation.

5.1 Model Performance Comparison

The models were first compared using precision, recall and F1-score, since these metrics are more informative than accuracy for imbalanced fraud detection problems.

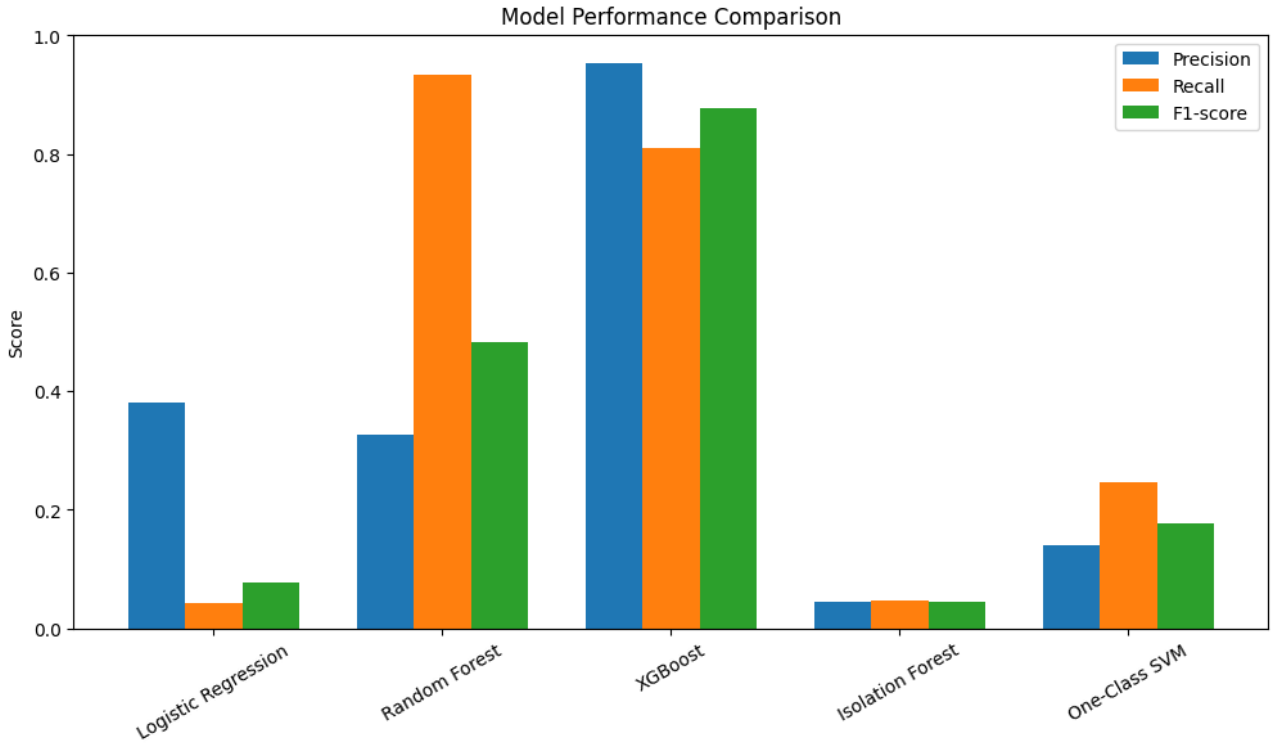

Figure 2: Model Performance Comparison Based on Precision, Recall, and F1-Score

The results show clear differences between the models. Logistic Regression performed poorly, especially in terms of recall, meaning that it failed to detect most fraudulent transactions. Isolation Forest and One-Class SVM also showed weak performance, indicating that unsupervised anomaly detection methods were not effective for this dataset.

Random Forest achieved very high recall, meaning it successfully detected most fraud cases. However, its precision was low, which means it produced many false positives. XGBoost achieved the best overall balance between precision and recall, resulting in the highest F1-score among the models. This indicates that XGBoost is the most balanced model from a machine learning performance perspective.

5.2 ROC-AUC Analysis

The ROC curve was used to compare the ability of the supervised models to distinguish between fraudulent and non-fraudulent transactions. This analysis was applied to Logistic Regression, Random Forest and XGBoost because these models provide probability outputs.

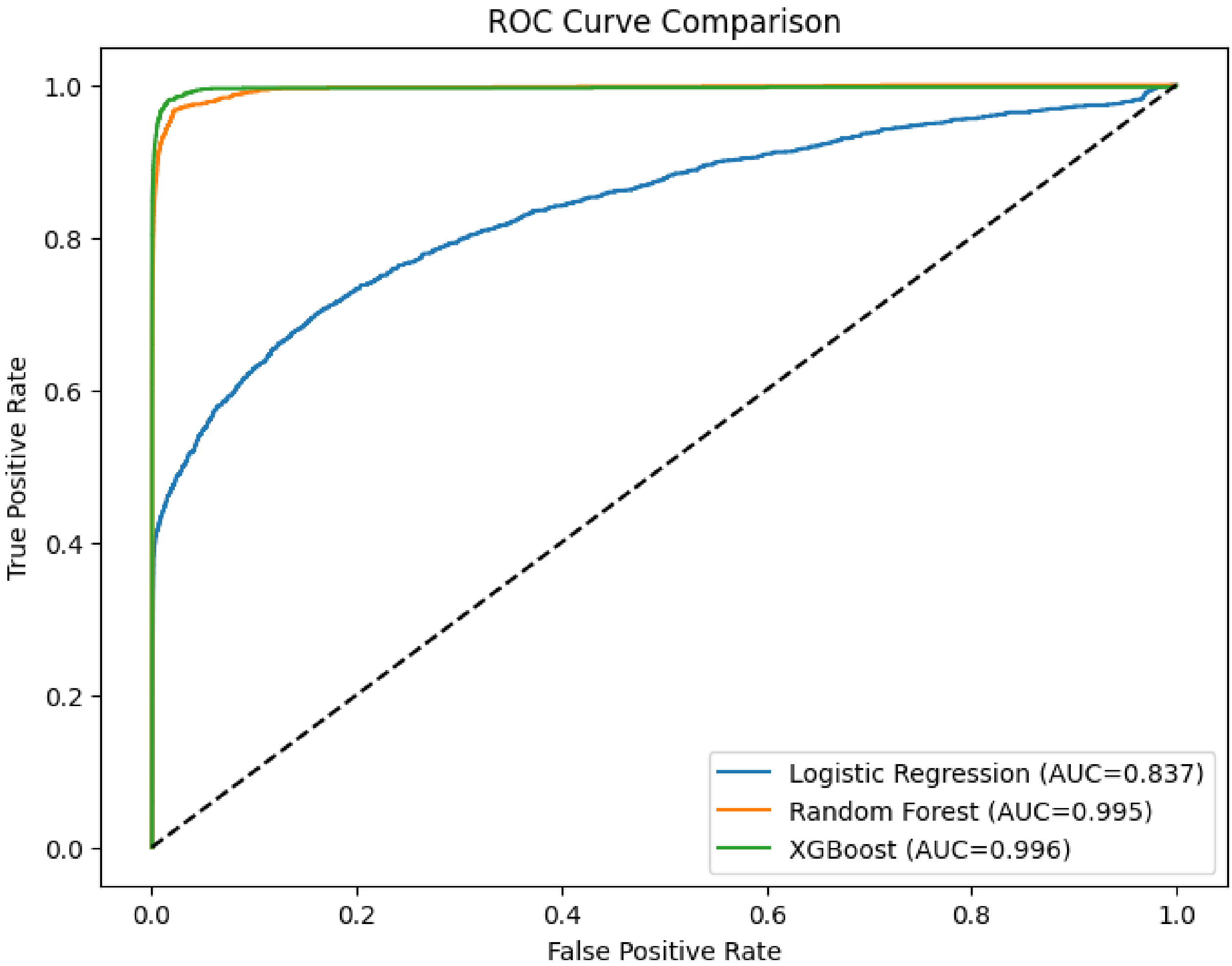

Figure 3: ROC Curve Comparison for Supervised Models

The ROC-AUC results show that Random Forest and XGBoost achieved very high scores, indicating strong discrimination ability between the two classes. XGBoost achieved the highest ROC-AUC score, confirming its strong overall classification performance. Logistic Regression achieved a lower ROC-AUC score, showing that it was less effective at separating fraudulent and non-fraudulent transactions.

5.3 Cross-Validation Results

To evaluate the stability of the supervised models, 5-fold Stratified Cross-Validation was applied. This method trains and tests the models on different subsets of the data while preserving the proportion of fraudulent and non-fraudulent transactions in each fold.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.9941 ± 0.001	0.3892 ± 0.0495	0.0477 ± 0.0068	0.0850 ± 0.0119	0.8348 ± 0.0110
XGBoost	0.9987 ± 0.0000	0.9511 ± 0.0044	0.8103 ± 0.0041	0.8750 ± 0.0026	0.9961 ± 0.0020

Table 1: 5-Fold Stratified Cross-Validation Results

To evaluate the stability of the supervised models, 5-fold Stratified Cross-Validation was applied. This method trains and tests the models on different subsets of the data while preserving the proportion of fraudulent and non-fraudulent transactions in each fold.

The cross-validation results confirm the findings obtained from the test set evaluation. Logistic Regression remained weak, with very low recall, meaning it consistently failed to detect most fraudulent transactions. Although its accuracy was high, this was mainly due to the strong class imbalance in the dataset.

XGBoost achieved consistently strong results across all folds, with high precision, recall, F1-score and ROC-AUC. The low standard deviation values show that the model performance was stable and not dependent on a specific train-test split. This confirms that XGBoost is a reliable model for fraud detection.

5.4 PCA Results

PCA was applied to evaluate whether dimensionality reduction could improve model performance. The comparison was performed on Logistic Regression and XGBoost by evaluating their results before and after applying PCA to the numerical features.

Model	PCA Applied	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	No	0.3810	0.0426	0.0767	0.8372
Logistic Regression	Yes	0.4035	0.0460	0.0825	0.8368
XGBoost	No	0.9538	0.8108	0.8765	0.9955
XGBoost	Yes	0.9010	0.6789	0.7743	0.9908

Table 2: Model Performance Before and After PCA

The PCA results show that dimensionality reduction did not significantly improve model performance. For Logistic Regression, PCA produced only a very small improvement in recall and F1-score, while the model still failed to detect most fraudulent transactions.

For XGBoost, PCA reduced performance, especially recall and F1-score. This indicates that some important fraud-related information may have been lost during dimensionality reduction. Therefore, PCA was not beneficial for this dataset, and the original feature space was more suitable for fraud detection.

5.5 Feature Importance Results

Feature importance analysis was performed using the XGBoost model because it achieved the best overall machine learning performance. This analysis helps identify which features contributed most to the model's fraud detection decisions.

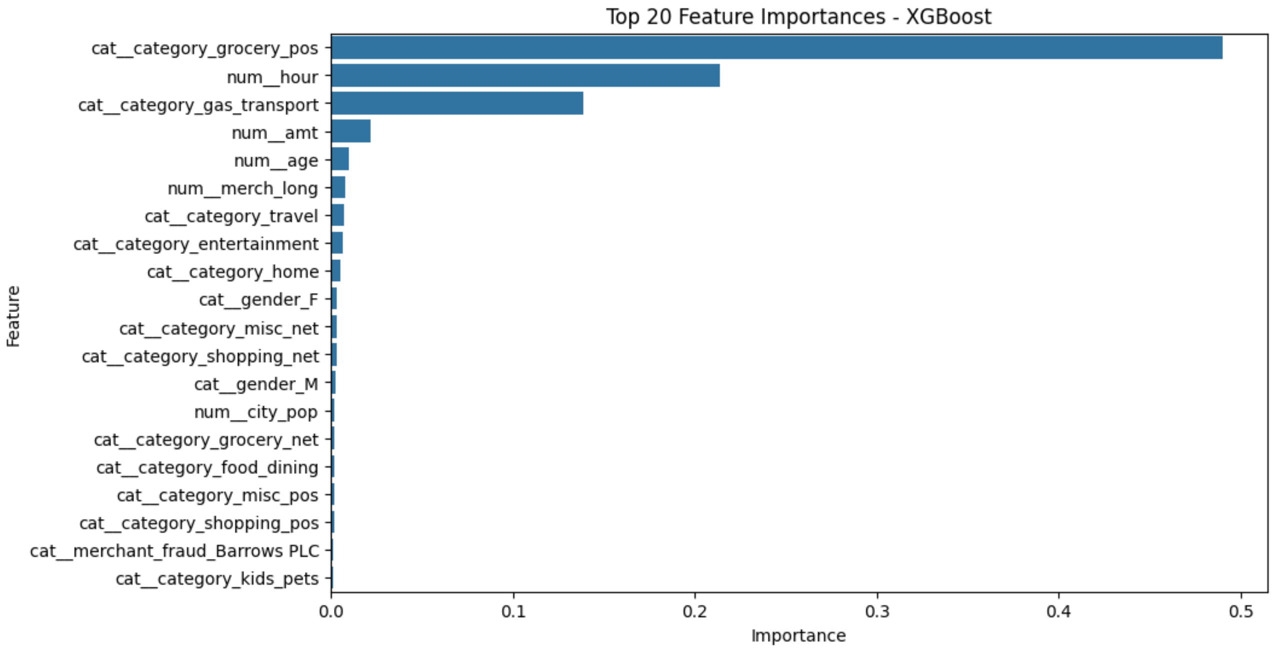

Figure 4: Top 20 Feature Importances for XGBoost

The feature importance results show that transaction category was the most influential factor in detecting fraud, especially categories such as grocery_pos and gas_transport. This indicates that certain types of transactions are more strongly associated with fraudulent behavior.

The transaction hour was also highly important, suggesting that time-based behavior plays a major role in fraud detection. This means that fraudulent transactions may occur more frequently during specific hours.

Interestingly, transaction amount was not the most important feature. This suggests that fraud detection depends more on behavioral patterns, transaction type and timing than on transaction size alone.

5.6 Business Cost Results

In addition to traditional evaluation metrics, the models were compared using a business cost-based approach. This evaluation assigns a higher cost to false negatives because missing a fraudulent transaction can lead to financial loss. False positives were assigned a lower cost because they mainly represent normal transactions that may require additional review.

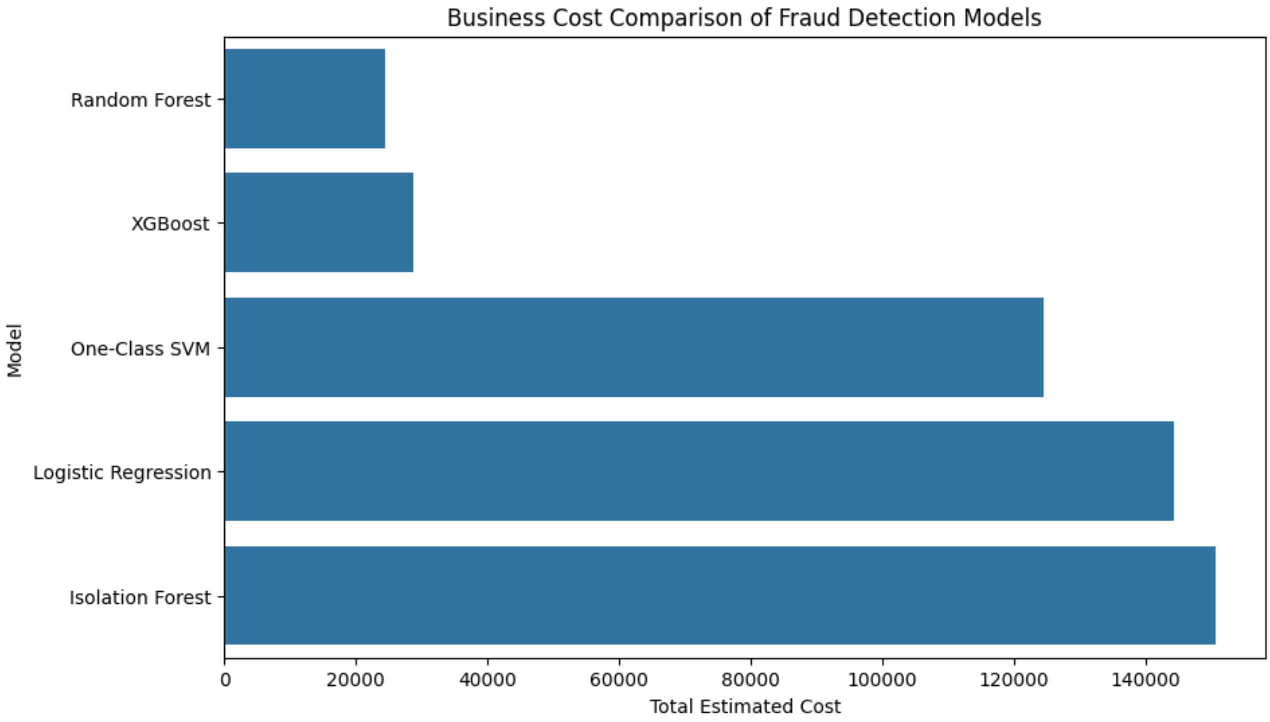

Figure 5: Business Cost Comparison of Fraud Detection Models

The business cost results show that Random Forest achieved the lowest total estimated cost. This is because it produced the fewest false negatives, meaning it missed fewer fraudulent transactions than the other models.

XGBoost achieved the second-lowest cost. Although XGBoost produced far fewer false positives than Random Forest, it missed more fraud cases. Since false negatives were assigned a higher cost, Random Forest became the preferred model from a business cost perspective.

This result highlights an important real-world insight: the best model depends on the objective. XGBoost is the best model based on traditional machine learning metrics, while Random Forest is more suitable when the main priority is reducing financial loss from missed fraud.

5.7 Final Model Selection

Based on the overall evaluation, XGBoost was selected as the best machine learning model from a statistical performance perspective. It achieved the best balance between precision and recall, the highest F1-score and the strongest ROC-AUC score among the models. This makes it the most reliable model when the goal is to detect fraud accurately while minimizing false alarms.

However, the business cost-based evaluation showed that Random Forest achieved the lowest estimated cost because it missed fewer fraudulent transactions. This means that Random Forest may be preferred in situations where the main objective is to reduce the financial loss cause by missed fraud cases.

Therefore, the final model choice depends on the objective of the fraud detection system. If the goal is to achieve the best overall classification performance, XGBoost is the most suitable model. If the goal is to minimize the cost of missed fraud, Random Forest may be more appropriate.

6 Scalability and Big Data Perspective

The dataset used in this project contains more than one million transaction records, which makes scalability an important consideration. Although the current implementation was completed using Python libraries such as pandas, scikit-learn and XGBoost, real-world financial institutions may process millions or billions of transactions every day.

For larger datasets, traditional in-memory processing may become inefficient or difficult to manage. In such cases, Big Data technologies such as Apache Spark can be used to distribute data processing and machine learning tasks across multiple machines. Spark can support scalable preprocessing, feature engineering and model training through distributed computation.

In a production fraud detection system, distributed storage systems such as HIDFS or cloud storage could be used to store large volumes of transaction data. Streaming technologies could also be used to process transactions in real time and detect fraud as soon as suspicious activity occurs.

Therefore, while the current implementation is suitable for this dataset, a real-world banking fraud detection system could benefit from Big Data tools to improve scalability, speed and real-time processing capability.

7 Conclusion

This project explored the use of machine learning techniques for fraud detection using a highly imbalanced transaction dataset. The main objective was to compare different supervised and unsupervised models and identify the most effective approach for detecting fraudulent transactions.

The results showed that simple models such as Logistic Regression were not sufficient for this task because they failed to detect most fraud cases. Unsupervised anomaly detection models, including Isolation Forest and One-Class SVM, also performed weakly compared to supervised models. This suggests that labeled fraud data is very useful for learning complex fraud patterns.

Among all models, XGBoost achieved the best overall machine learning performance. It provided a strong balance between precision and recall, achieved the highest F1-score and showed excellent

ROC-AUC performance. Cross-validation confirmed that its results were stable and reliable across different data splits.

The PCA experiment showed that dimensionality reduction did not improve model performance. This indicates that the original features contained important information for fraud detection and that reducing the feature space caused some useful patterns to be lost.

Feature importance analysis showed that transaction category and transaction time were among the most influential indicators of fraud. This suggests that fraud detection depends strongly on behavioral patterns, not only on transaction amount.

Finally, the business cost-based evaluation showed that Random Forest achieved the lowest estimated cost because it missed fewer fraud cases. This highlighted an important practical insight: the best model depends on the objective. XGBoost is the best choice for balanced machine learning performance, while Random Forest may be preferred when the main goal is minimizing the financial cost of missed fraud.

Overall, this project demonstrates that advanced supervised learning models, combined with proper preprocessing, evaluation metrics, interpretability and business-oriented analysis, can provide effective solutions for fraud detection.

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9 Annexes

Annex A: Confusion Matrices

This annex presents additional confusion matrices for the main supervised models. These matrices provide more detail about the number of correct and incorrect predictions made by each model.

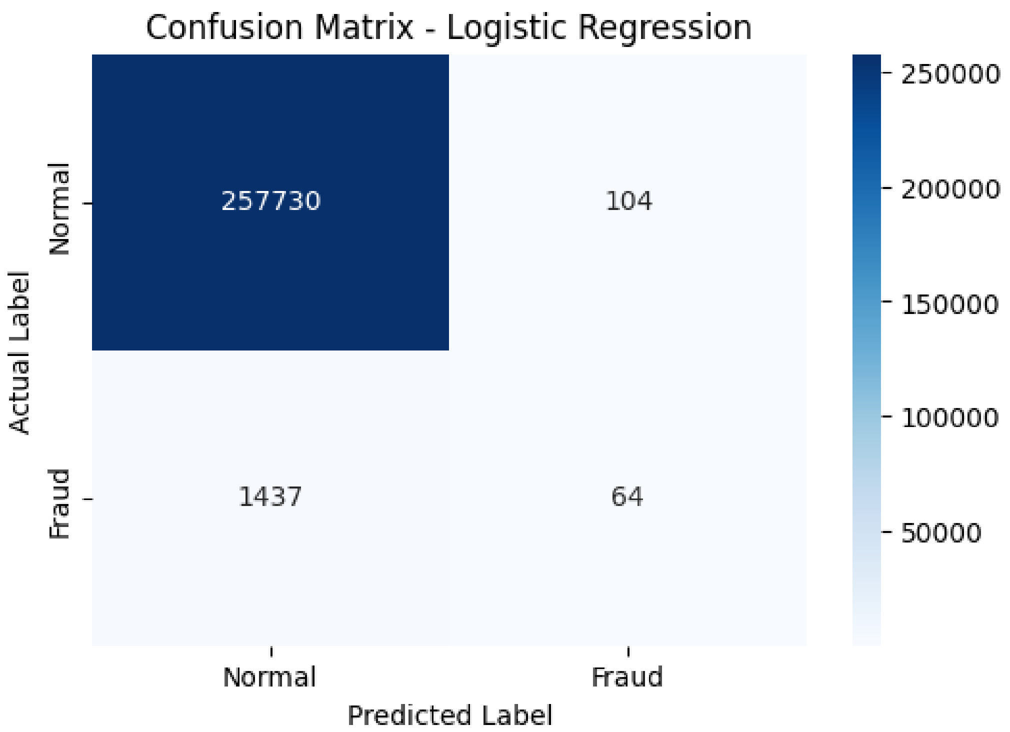

Figure A1: Confusion Matrix for Logistic Regression

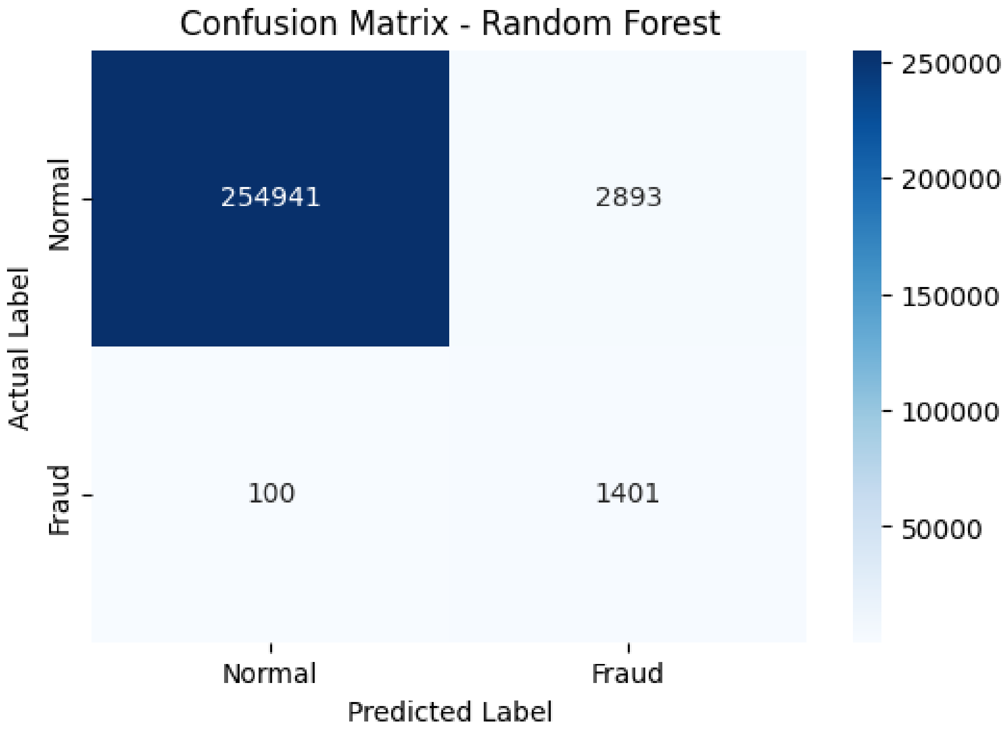

Figure A2: Confusion Matrix for Random Forest

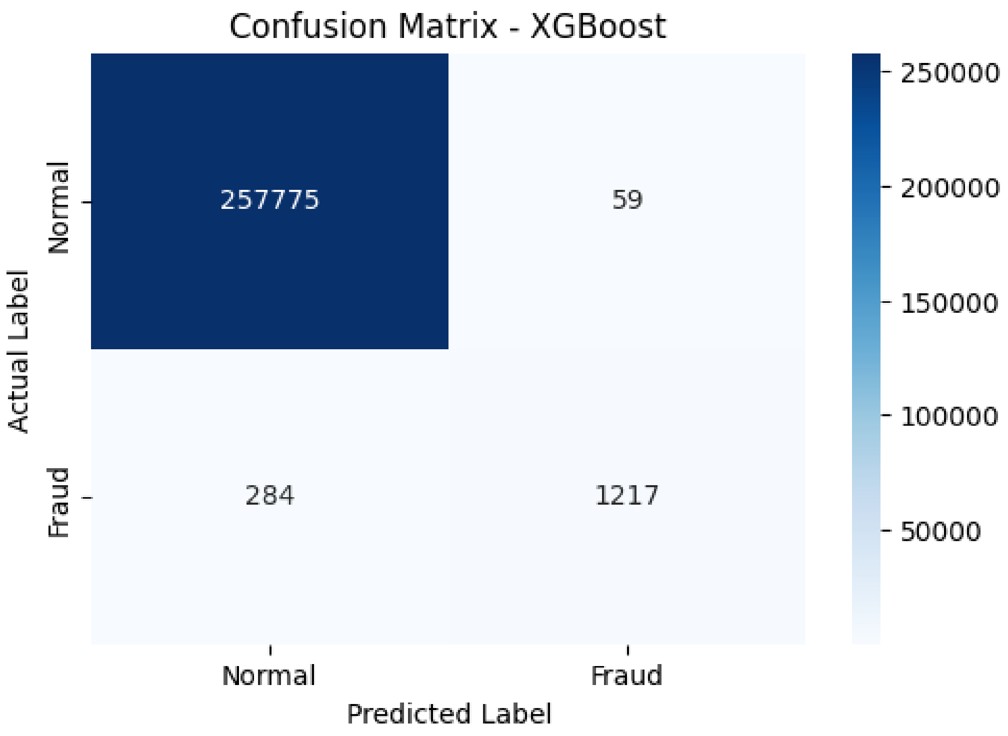

Figure A3: Confusion Matrix for XGBoost

Annex B: Additional Model Evaluation Results

The table below summarizes the main evaluation metrics for all implemented models. These results support the model comparison discussed in the main report.

Model	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.3810	0.0426	0.0767	0.8372
Random Forest	0.3263	0.9334	0.4835	0.9948
XGBoost	0.9538	0.8108	0.8765	0.9955
Isolation Forest	0.0442	0.0473	0.0457	Not applied
One-Class SVM	0.1393	0.2465	0.1780	Not applied

Table A1: Additional Evaluation Metrics for All Models

Annex C: Notebook Implementation

The complete implementation of this project is provided in the accompanying Jupyter Notebook. The notebook includes data preprocessing, feature engineering, model training, model evaluation, cross-validation, PCA implementation, feature importance analysis, visualizations and business cost-based evaluation.

The notebook was used as the main technical implementation file, while this report summarizes the methodology, results and main findings of the project.